

Case Study 1: Medical Department Demand and Resource Allocation Analysis

Explanation

This case study analyzes the number of patient consultations across different medical departments to identify the most demanded healthcare services.

Aim

To determine which departments receive the highest consultation volume and support healthcare resource allocation.

Input

Department_Label*, Question_Sequence, ID, Timestamp

Techniques / Functions

Filtering, Grouping, Counting, Sorting, Retrieval

Output

A ranked list of medical departments based on consultation frequency, including the top high-demand departments.

Key = Department_Label

Value = Total_Number_of_Consultations

Case Study 2: Disease and Symptom Trend Mining

Explanation

This case study extracts frequent disease and symptom patterns from patient questions, expert answers, and related disease labels.

Aim

To identify common medical concerns and disease trends within large-scale consultation data.

Input

Question_Sequence, Answer_Sequence, Related_Diseases*,
Department_Label

Techniques / Functions

WordCount, Pattern Matching, Grouping, Counting, Sorting

Output

A ranked frequency summary of common diseases, symptoms, and disease-related keywords.

Case Study 3: Expert Advice Quality and Hospital Performance Analysis

Explanation

This case study evaluates expert response quality by analyzing expert scores across hospitals, departments, and doctors.

Aim

To compare the consistency and quality of medical advice across different healthcare providers.

Input

Expert_Score, Hospital_Org*, Doctor_Name, Department_Label,
Answer_Sequence

Techniques / Functions

Average/Mean Calculation, Partitioning, Filtering, Min/Max Retrieval, Sorting

Output

A comparative report showing average expert scores, highest-rated providers, lowest-rated records, and quality variations across hospitals or departments.

Case Study 4: Temporal Medical Consultation Trend and Framework Performance Analysis

Explanation

This case study analyzes how consultation volume and disease-related questions change over time, while also comparing Hadoop and Spark performance using the same processing tasks.

Aim

To identify time-based consultation trends and evaluate which big data framework processes the dataset more efficiently.

Input

Timestamp*, Question_Sequence, Related_Diseases, Department_Label, full
Huatuo-26M dataset

Techniques / Functions

Timestamp Extraction, Grouping, Aggregation/Counting, WordCount, Processing-Time Calculation

Output

A time-based trend report showing consultation patterns by day, week, or month, together with a Hadoop vs. Spark processing-time comparison.